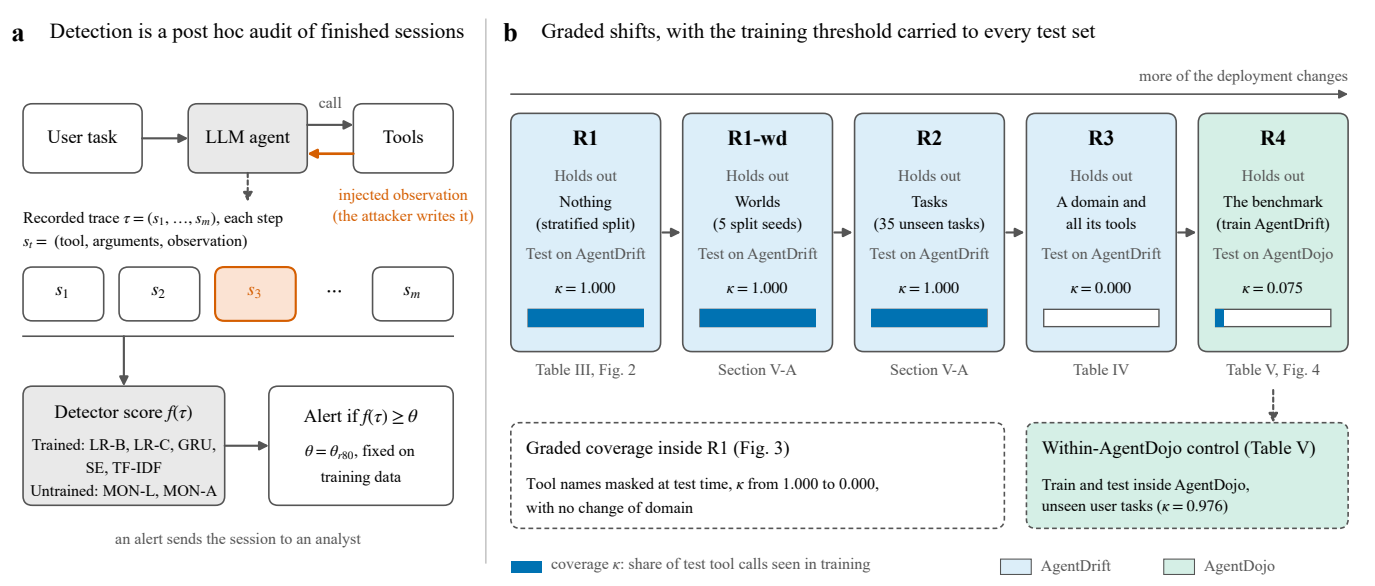

a
Detection is a post hoc audit of finished sessions
User task
LLM agent
call
Tools
injected observation
(the attacker writes it)
Recorded trace $\tau = (s_1, \ldots, s_m)$, each step
$s_t =$ (tool, arguments, observation)
s_1
s_2
s_3
...
s_m
Detector score $f(\tau)$
Trained: LR-B, LR-C, GRU, SE, TF-IDF
Untrained: MON-L, MON-A
Alert if $f(\tau) \geq \theta$
$\theta = \theta_{r80}$, fixed on training data
an alert sends the session to an analyst
b
Graded shifts, with the training threshold carried to every test set
more of the deployment changes
R1
Holds out
Nothing
(stratified split)
Test on AgentDrift
$\kappa = 1.000$
Table III, Fig. 2
R1-wd
Holds out
Worlds
(5 split seeds)
Test on AgentDrift
$\kappa = 1.000$
Section V-A
R2
Holds out
Tasks
(35 unseen tasks)
Test on AgentDrift
$\kappa = 1.000$
Section V-A
R3
Holds out
A domain and
all its tools
Test on AgentDrift
$\kappa = 0.000$
Table IV
R4
Holds out
The benchmark
(train AgentDrift)
Test on AgentDojo
$\kappa = 0.075$
Table V, Fig. 4
Graded coverage inside R1 (Fig. 3)
Tool names masked at test time, κ from 1.000 to 0.000,
with no change of domain
Within-AgentDojo control (Table V)
Train and test inside AgentDojo,
unseen user tasks ($\kappa = 0.976$)
coverage κ: share of test tool calls seen in training
AgentDrift
AgentDojo